Rich6D simulation closeout

12 October 2026

I. Simulations: a constant calibrated once

We use the frozen Rich6D economic experiment: a six-dimensional Matérn-3/2 representation, Fourier truncation $M = 128$, and 300 independent histories in each of five paired economies. The baseline has $N = 600$ and $\rho = 0.95$; alternatives change N to 300 or 1200, or persistence to 0 or 0.75. Analytic economic optima are specific to N. Sharpe ratios are marginal monthly population quantities multiplied by $\sqrt{12}$; response-one loss is not annualized.

For each economy and history, initial cross-validation selects one constant from the unchanged 71-point grid $[10^{-8}, 10^{-1}]$ using observations 1–60 only. Four forward folds train on 20, 30, 40, and 50 observations and validate on the subsequent ten. Each fold uses $\lambda = an_{\text{train}}^{-0.6}$ and observation-weighted response-one loss; exact ties favor larger a. The selected constant is then frozen:

$$\lambda_{k,T} = \widehat{a}_{k,60} T^{-0.6}.$$

Later observations refit the policy, with no further calibration. The historical constant is a comparison. Oracle Q minimizes mean regret and oracle SR maximizes mean Sharpe on the original 96-penalty grid, choosing one common penalty per economy and horizon. Both use population information and are diagnostics only.

Table I. Baseline endpoint at $T = 4860$, all 300 histories.

Specification	Annual SR	Regret	Empirical C
Initial CV	2.93250	0.01130	143.78233
Historical fixed	2.93477	0.02220	14.95632
Oracle Q	2.95181	0.00833	58.32002
Oracle SR	2.95343	0.00866	41.74288

Initial CV reduces baseline mean endpoint regret from 0.022205 to 0.011296 (paired change -0.010909, 95% Monte Carlo mean interval $[-0.011399, -0.010419]$). Annualized Sharpe changes from 2.934774 to 2.932497 (paired change -0.002277, interval $[-0.005079, 0.000525]$). Empirical complexity rises from 14.956 to 143.782. Thus the response-one improvement does not establish a final Sharpe improvement. Oracle Q and oracle SR yield distinct endpoint penalties and outcomes; they do not tune the strategy.

The financial components uniquely identified from each policy's signed Sharpe and regret explain this mean discrepancy. At $T = 4860$, mean positive-ray loss is 0.011119 under initial CV versus 0.010733 historically, while mean scale loss falls from 0.011472 to 0.000177. These add to the respective mean regrets. Thus improved scaling more than offsets worse directional fit in the response-one objective. These quantities require no choice of a payoff-moment root; actual monthly moments and the remaining radial decomposition are recovered separately on the audit subset.

The full-grid descriptive slopes of log mean regret and log mean Sharpe gap are -0.437820 and -0.398648, respectively. The unchanged windows $T \geq 360$, $T \geq 720$, and $T \geq 1440$, full cross-horizon covariance MCSE, jackknife checks, and normalized-outcome quantiles are reported in the appendix. No horizon or replication was removed to approach -0.6. The asymptotic statement is an $O_{\mathbb{P}}$ upper bound, not an equality for a finite Monte Carlo mean.

The ideal infinite Fourier target, the simulated $M = 128$ economy, and the fixed-dimensional Nyström estimator are distinct objects. The attached derivation supports maximal source order $r = 1$ for the ideal target and preserves the kernel upper rate under a frozen selection on a finite positive grid, conditional on the manuscript assumptions. It does not certify the numerical approximation. A fixed subspace with positive representation error cannot have regret to the full economic optimum converging to zero. At the baseline endpoint the rank-4096 representation floor is about 15.7% of mean CV regret, exceeding the unchanged 5% criterion. The paired CV rank and quadrature audit remains unresolved; its scope and failed cells are tabulated in the appendix.

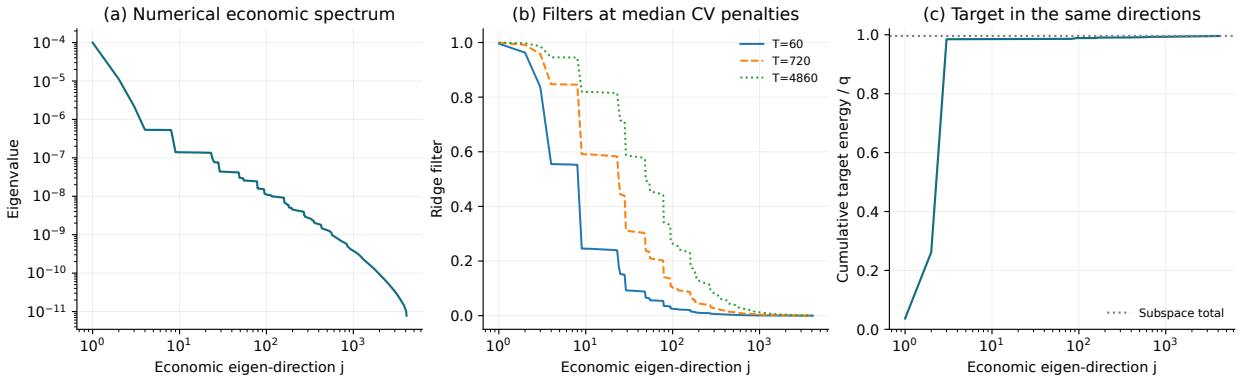


Figure 1. Economic spectrum, ridge filters at median initial-CV penalties, and target energy measured in the same numerical eigen-directions. The displayed operator is the frozen rank-4096 approximation; it is not an infinite-dimensional spectrum.

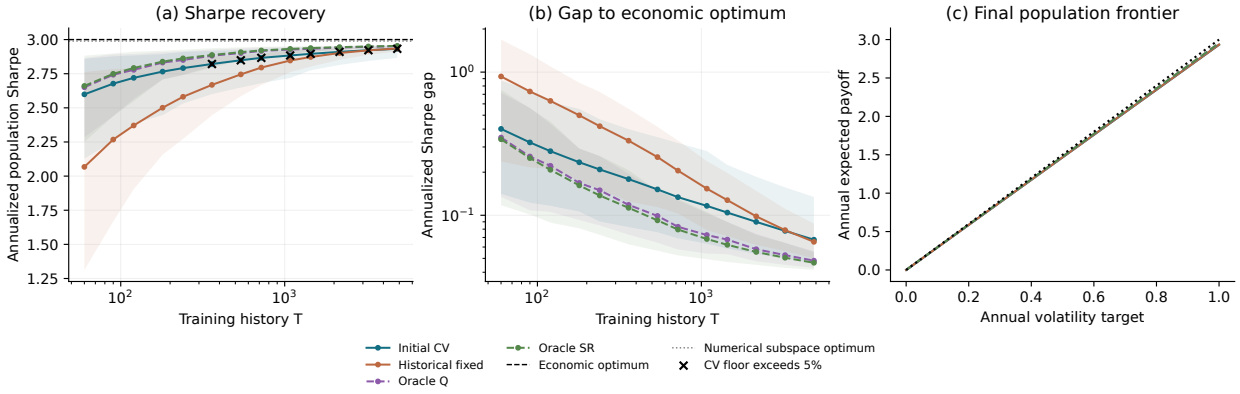


Figure 2. Baseline population Sharpe recovery, gap to the analytic economic optimum, and final capital-allocation frontiers. Curves include every archived horizon. Shaded bands are the central 95 percent of replication outcomes, not confidence intervals of the mean; oracle bands are descriptive and not selection adjusted. Crosses mark CV projection floor/mean regret exceeding 5 percent. The CV audit remains scope-limited; rank and quadrature uncertainty are additional.

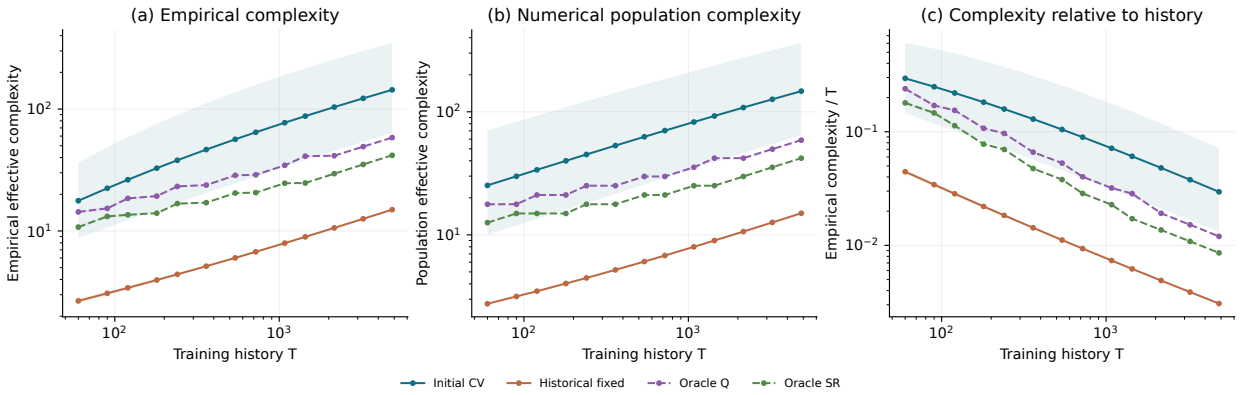


Figure 3. Empirical and numerical population effective complexity, and empirical complexity divided by history length. Population complexity is evaluated at each actual selected penalty before aggregation; it is not complexity at the mean penalty. Bands are replication percentiles.

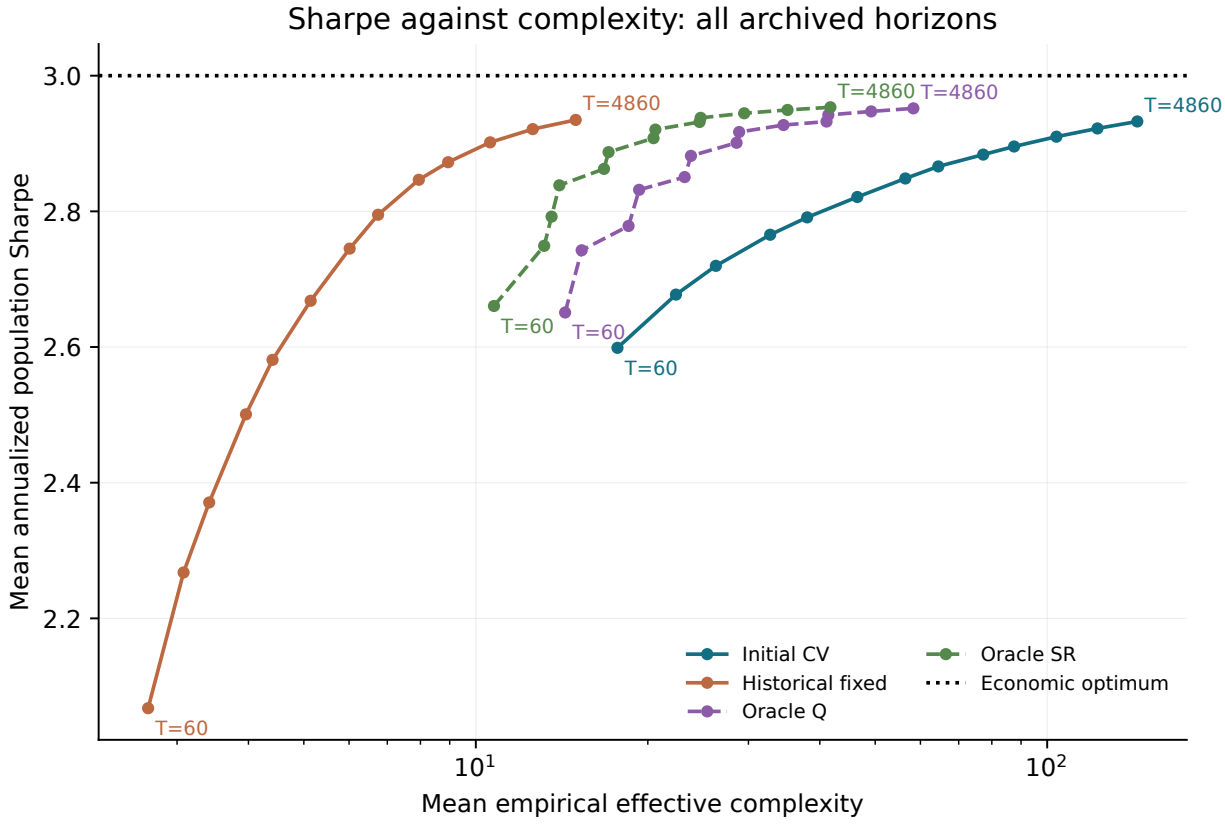


Figure 4. Mean population Sharpe against mean empirical complexity across the full baseline horizon grid. Solid paths implement frozen initial CV or the historical rule; dashed oracle paths use population information. The horizontal line is the true economic optimum.

A. Simulation diagnostics and numerical scope

All diagnostics added during closeout are analyses after observing the archived experiment. The economic specification, folds, grid, pairing, initial calibration time, exponent and all original outcomes are immutable. Full per-history data, normalized outcomes, distribution summaries and reproduction commands accompany the generated tables.

Table II. Initial-CV descriptive baseline slopes on the established windows. The slope is the regression of log Monte Carlo mean, not the mean of per-history slopes. Full covariance preserves complete history dependence across horizons.

Quantity	Window	Horizons	Slope	MCSE	Jackknife
Regret	full	13	-0.43782	0.00556	0.00557
Regret	T_ge_360	8	-0.40437	0.00541	0.00542
Regret	T_ge_720	6	-0.38727	0.00636	0.00637
Regret	T_ge_1440	4	-0.38125	0.00740	0.00742
Sharpe gap	full	13	-0.39865	0.00552	0.00553
Sharpe gap	T_ge_360	8	-0.37208	0.00486	0.00486
Sharpe gap	T_ge_720	6	-0.36043	0.00585	0.00586
Sharpe gap	T_ge_1440	4	-0.35854	0.00706	0.00707

Table III. Initial-CV endpoints in the five economies, all 300 histories. The common comparison grid ends at $T = 3240$; baseline continues to 4860.

Economy	T	Annual SR	Regret
N1200	3240	2.97420	0.00877
N300	3240	2.81517	0.02529
baseline	4860	2.93250	0.01130
rho000	3240	2.91578	0.01538
rho075	3240	2.91842	0.01463

For each policy, let $u = \mathbb{E}Y$, $v = \mathbb{E}Y^2$ and $q = 1 - Q(W^\star)$. These are monthly moments. Set $c = u/q$, $d = v - u^2/q$. Direct calculation verifies

$$Q(W) - Q(W^\star) = q(c-1)^2 + d, \qquad \mathrm{SR}(W) = \frac{u}{\sqrt{v - u^2}}.$$

For $c > 0$, the scale-invariant directional error is $d/[c^2 q(1-q)]$. For $u, v > 0$, the positive-ray identity is

$$\underbrace{q - u^2/v}_{\text{ray gap}} + \underbrace{(v-u)^2/v}_{\text{scale gap}} = Q(W) - Q(W^\star).$$

Null or negative means and zero variances are recorded separately, without changing policy orientation. Roundoff-sized negative directional errors are retained; larger negative errors fail verification. Nonlinear identities are checked per policy rather than on aggregate moments.

For the same numerical subspace and actual penalty, let $\beta_P = S^{-1}m$, $\beta_\lambda = (S + \lambda I)^{-1}m$, and $\Delta = \widehat{\beta} - \beta_\lambda$. Then

$$\begin{aligned} Q(\widehat{\beta}) - Q(W^\star) = q - m'S^{-1}m + (\beta_\lambda - \beta_P)'S(\beta_\lambda - \beta_P) \\ + \Delta'S\Delta + 2\Delta'S(\beta_\lambda - \beta_P). \end{aligned}$$

The signed cross term is preserved. The displacement norm is not automatically a variance, particularly with a data-selected penalty. Recovered-moment tables explicitly state the number of histories and horizons re-estimated; they must not be read as decompositions of all 1500 archived histories.

Table IV. Baseline endpoint financial components, all 300 histories. Ray/scale quantities are uniquely identified from positive signed Sharpe and regret for each policy. Population-ridge bias uses each actual penalty in the same numerical operator. The last column keeps displacement and signed cross combined; it is not variance. Their separate values require recovered coefficients.

Specification	Ray	Scale	Floor	Bias	Displacement + cross
Initial CV	0.01112	0.00018	0.00178	0.00318	0.00634
Historical	0.01073	0.01147	0.00178	0.01934	0.00109

The numerical protocol separates A (the archived rank-4096 constant held fixed at both ranks) from B (initial CV repeated only on the identical first 60 observations at each rank). It declares twelve representative original production indices and separate stress indices selected by the smallest initial constant. These differ from the historical rank-pilot seeds. Full and short audits have separate

Table V. Actual recovered monthly moments and radial/angular components, baseline T=4860. Scope is the same explicitly counted audit histories as the following table; these values are not inferred by choosing roots of Q/SR.

Policy	u	v	c	d	Radial	Normalized direction	Monthly SR
A	0.429376	0.440028	1.001877	0.009846	0.000002	0.040055	0.849185
C	0.362138	0.315970	0.844989	0.009968	0.010298	0.057004	0.842350

Table VI. Recovered baseline endpoint moments. A uses frozen initial CV; C uses the historical constant. Scope: A: 1 histories, C: 1 histories; descriptive paired audit histories only. Ray plus scale and floor plus bias plus displacement plus signed cross reconstruct regret.

Policy	Ray	Scale	Floor	Bias	Displacement	Cross
A	0.00959	0.00026	0.00178	0.00368	0.00437	0.00002
C	0.01352	0.00675	0.00178	0.01934	0.00116	-0.00201

labels. The original 5% projection-floor criterion, paired absolute mean plus 1.96 MCSE relative to mean rank-4096 regret, and independent-quadrature comparison remain unchanged. Mean, second moment, Sharpe-gap and normalized-directional sensitivity thresholds are also 5%. No subset result certifies an untested horizon or provides a forward certificate for rank 8192.

The new CV audit re-estimates full audit horizons for 1 original production histories and only T=60 for 15 additional histories. There are 0 failed numerical metric comparisons and 450 comparisons with insufficient independent histories for MCSE. Rank-specific initial CV changes 8 of 80 checked selections. Representative-history MCSE is computed separately from minimum-a stress diagnostics; stress cases are never pooled into an IID MCSE. The twelve-history full-horizon protocol and separate stress coverage are incomplete; uncertainty at untested cells remains unresolved. Rank 8192 has no forward-rank certificate. Population-ridge decomposition at rank 8192 has not run. Quadrature differences are systematic evaluation diagnostics; replication MCSE does not quantify uncertainty over quadrature seeds.

The infinite Fourier tail satisfies $\|\psi_\infty - \psi_M\|_\infty \le c_\psi/[4M^4 \log(M+1)]$, about 6.6991×10^{-9} at $M = 128$. For fixed policies this implies uniform loading and moment perturbation bounds. The

Table VII. Paired rank8192 minus4096 endpoint changes on the recovered audit histories. A holds the archived constant fixed; B reselects once on the initial60 at each rank. These are point differences. Where n=1 a replication MCSE is not estimable; the complete representative/stress endpoint audit remains unresolved.

Economy	T	Audit	n	ΔQ	Δu	Δv	Δ SR gap	Δ direction
N1200	3240	A	1	3.669e-06	1.660e-05	3.688e-05	1.803e-05	1.235e-05
N1200	3240	B	1	3.669e-06	1.660e-05	3.688e-05	1.803e-05	1.235e-05
N300	3240	A	1	4.284e-05	7.079e-05	1.844e-04	2.051e-04	1.542e-04
N300	3240	B	1	4.284e-05	7.079e-05	1.844e-04	2.051e-04	1.542e-04
baseline	4860	A	1	2.468e-06	2.421e-05	5.089e-05	7.278e-06	5.146e-06
baseline	4860	B	1	2.468e-06	2.421e-05	5.089e-05	7.278e-06	5.146e-06
rho000	3240	A	1	-1.186e-06	6.078e-05	1.204e-04	-8.762e-06	-6.178e-06
rho000	3240	B	1	-1.186e-06	6.078e-05	1.204e-04	-8.762e-06	-6.178e-06
rho075	3240	A	1	6.156e-06	4.861e-05	1.034e-04	2.956e-05	2.096e-05
rho075	3240	B	1	6.156e-06	4.861e-05	1.034e-04	2.956e-05	2.096e-05

exported operator perturbation ratios use every actually selected penalty. Small pointwise error alone does not establish stability of a ridge inverse or an initial argmin; paired Fourier diagnostics and any remaining gaps are explicitly separated. The finite truncation does not identify a maximal source order from its differentiability.

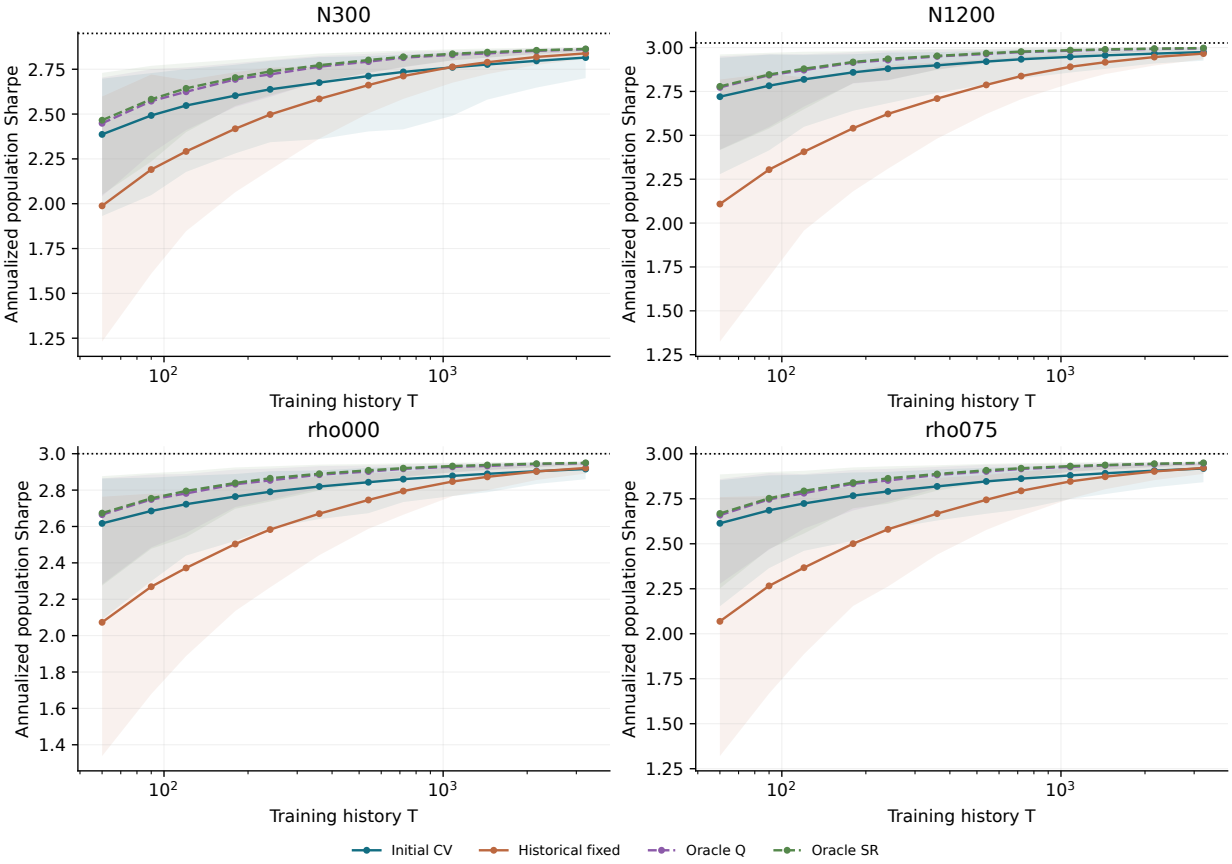


Figure 5. All four alternative economies on the common grid, with their analytic economic optima. Bands show replication outcomes. Economies differing only in persistence share the marginal population operator but can select different penalties and complexities.

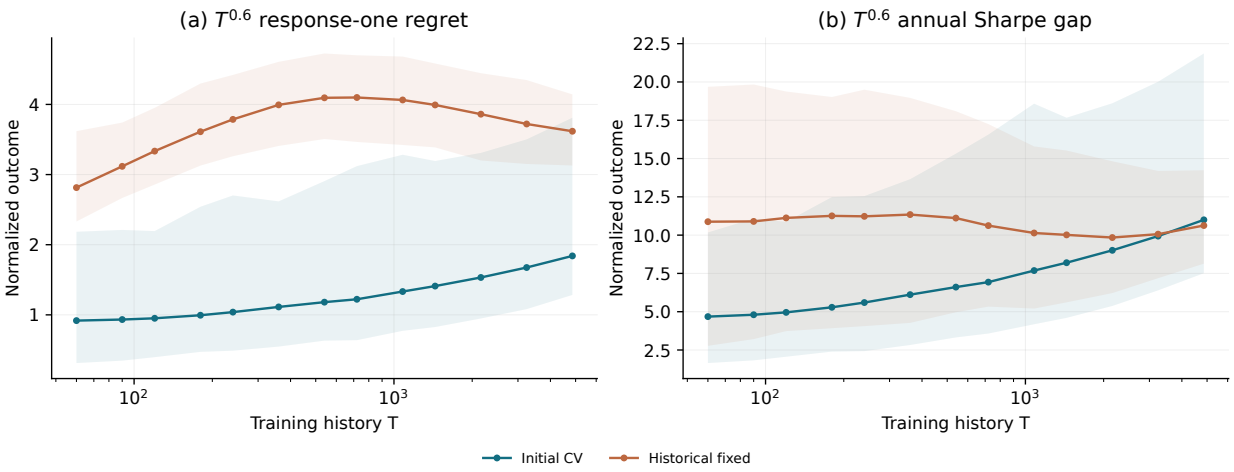


Figure 6. Normalized baseline regret and Sharpe gap on the complete grid. Means and replication-percentile bands are finite descriptive distributions; they do not identify an asymptotic rate in probability.